

Retail Demand Forecasting

Project Report

Abstract

Retail demand forecasting plays a significant role in helping businesses estimate future product demand and maintain optimal inventory levels. This project focuses on forecasting retail sales using the **M5 Forecasting Dataset** and the **Facebook Prophet** time-series forecasting model. Historical sales data was analyzed, cleaned, and visualized to identify sales patterns and trends. The trained forecasting model was used to generate a **30-day sales forecast**, and the results were presented through an interactive **Streamlit dashboard**. The dashboard provides forecast data, sales summaries, inventory recommendations, dataset information, and model information. This project demonstrates how forecasting techniques can support inventory planning and improve business decision-making.

Introduction

Retail businesses need accurate demand forecasts to ensure products are available when customers require them. Incorrect demand estimation may lead to either excess inventory or stock shortages, affecting both customer satisfaction and business profitability.

Demand forecasting is a data-driven approach that predicts future sales based on historical information. In this project, the **Facebook Prophet** forecasting model was used to analyze historical retail sales data and predict product demand for the next 30 days. Before model development, the dataset was cleaned and explored through various visualizations to understand sales patterns, pricing trends, and seasonal behavior.

An interactive dashboard was developed using **Streamlit** to display forecast results, sales summaries, inventory recommendations, dataset information, and model details. This dashboard provides an easy-to-use interface for understanding future sales trends and supports inventory planning and business decision-making.

Problem Statement

Retail businesses often face challenges in maintaining optimal inventory levels due to fluctuations in customer demand. Overstocking increases storage costs and product wastage, while understocking leads to lost sales and dissatisfied customers. Traditional forecasting methods may not accurately capture seasonal variations and changing demand patterns. Therefore, an efficient forecasting system is required to analyze historical sales data and generate reliable demand predictions. This project addresses these challenges by using the Facebook Prophet model to forecast retail demand and support inventory planning.

Objectives

The main objectives of this project are:

- Analyze historical retail sales data.
 - Perform data cleaning and preprocessing.
 - Explore sales patterns using data visualization.
 - Forecast retail demand for the next 30 days.
 - Develop an interactive Streamlit dashboard.
 - Support inventory planning and business decision-making.
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Dataset Description

The project uses the **M5 Forecasting Dataset**, which is a publicly available dataset for retail demand forecasting. It contains historical sales data, calendar information, and selling prices for products across multiple stores. The dataset is widely used for time-series forecasting and demand prediction.

Tools & Technologies

Programming Language

- Python

Libraries

- Pandas
- NumPy
- Matplotlib
- Facebook Prophet
- Streamlit

Development Tools

- Jupyter Notebook

System Requirements

Hardware Requirements

- Intel Core i3 or higher
- 8 GB RAM
- 20 GB Free Storage

Software Requirements

- Windows 10/11
- Python 3.x
- Jupyter Notebook
- Streamlit

Methodology

The project was developed using the following methodology:

1.Data Collection

The M5 Forecasting Dataset was used as the source of historical retail sales data. The dataset contains product sales, calendar information, and selling prices, which were used for demand forecasting.

2. Data Cleaning

The dataset was cleaned by checking for missing values, removing unnecessary columns, and converting data into a suitable format for analysis. This ensured the quality and consistency of the data before modeling.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the sales patterns and characteristics of the dataset. Various charts and graphs were created to analyze product sales, pricing trends, monthly distribution, weekday distribution, and event types.

4. Feature Preparation

The data was prepared for forecasting by selecting the required columns and formatting them according to the Facebook Prophet model. The date column was renamed to **ds**, and the sales column was renamed to **y**, as required by Prophet.

5. Model Building Using Facebook Prophet

The Facebook Prophet model was trained using the prepared historical sales data. Prophet automatically identifies trends and seasonal patterns, making it suitable for retail demand forecasting.

6. Forecast Generation

After training the model, a **30-day sales forecast** was generated. The predicted sales values were stored and visualized to understand future demand trends.

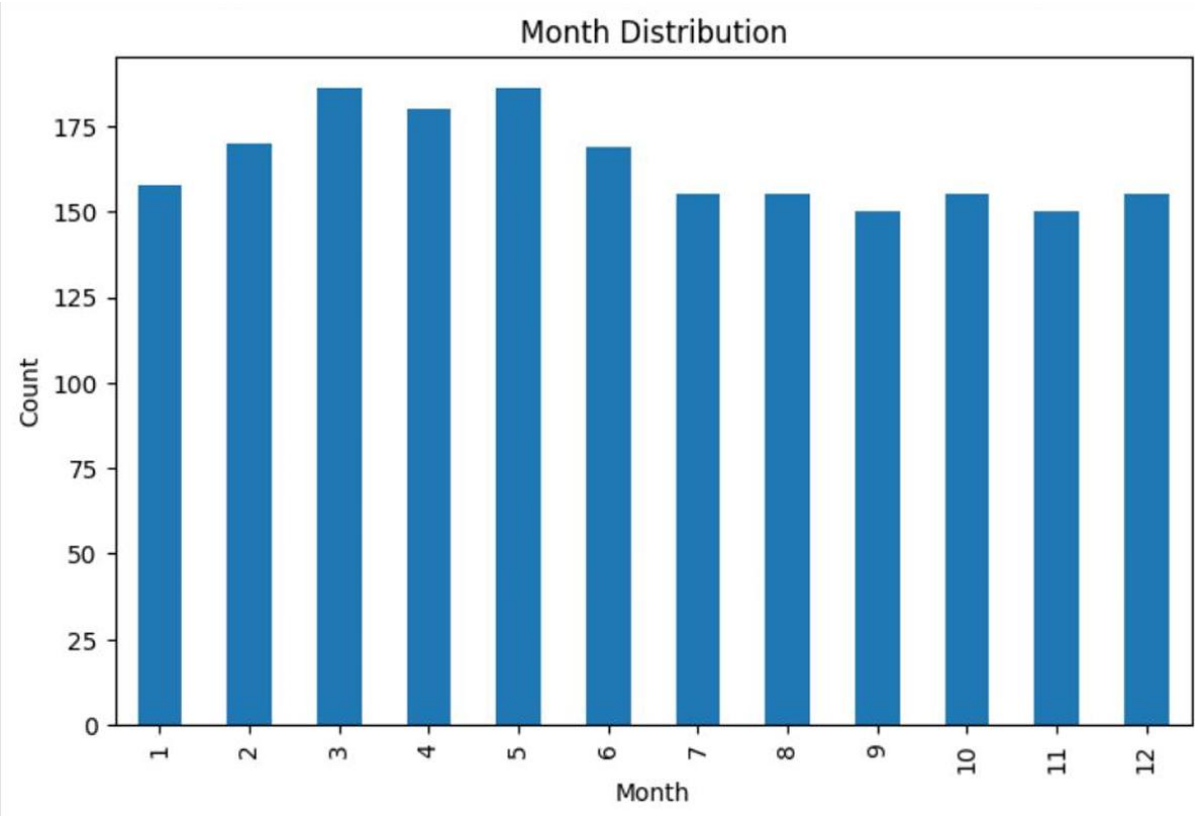
7. Streamlit Dashboard Development

An interactive dashboard was developed using Streamlit to display the forecast results. The dashboard includes sales forecasts, sales summary, inventory recommendations, dataset information, model information, and various visualizations to help users interpret the results easily.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the retail sales dataset before building the forecasting model. Various visualizations were created to identify trends, seasonal patterns, pricing behavior, and product distribution. These insights helped in understanding the data and preparing it for accurate demand forecasting.

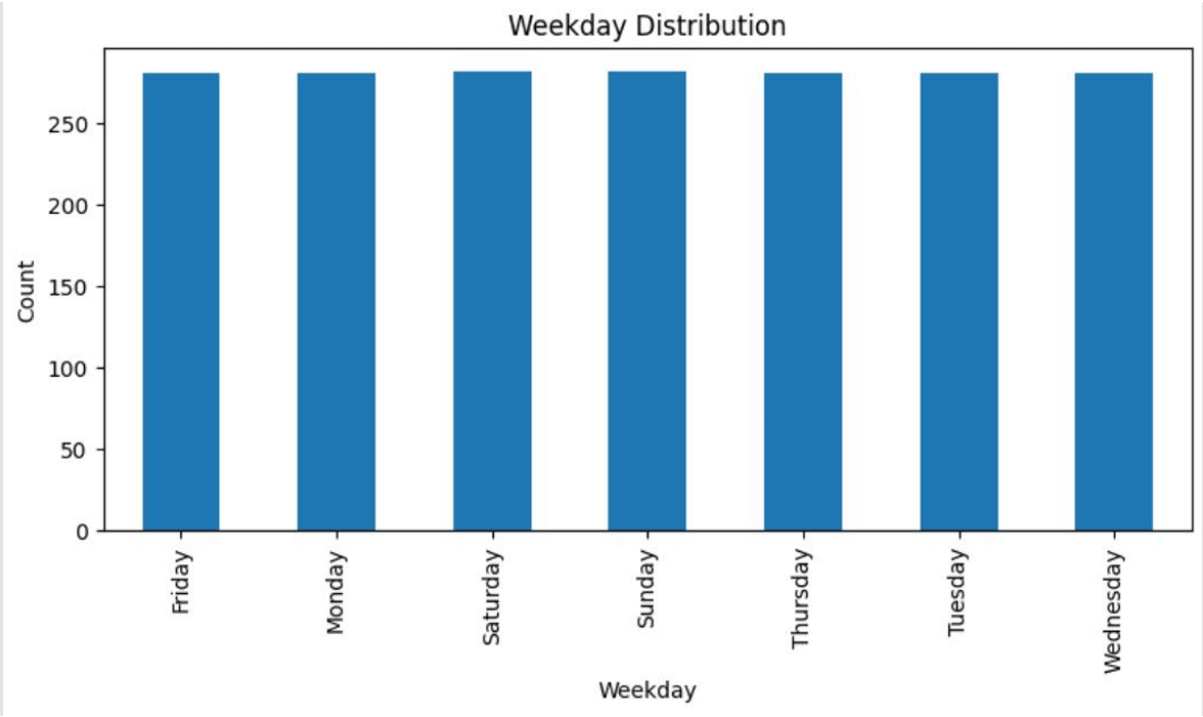
1. Month Distribution



Explanation:

This chart shows the distribution of sales across different months. It helps identify monthly sales patterns and seasonal variations in customer demand. Understanding these trends is useful for predicting future sales and planning inventory more effectively.

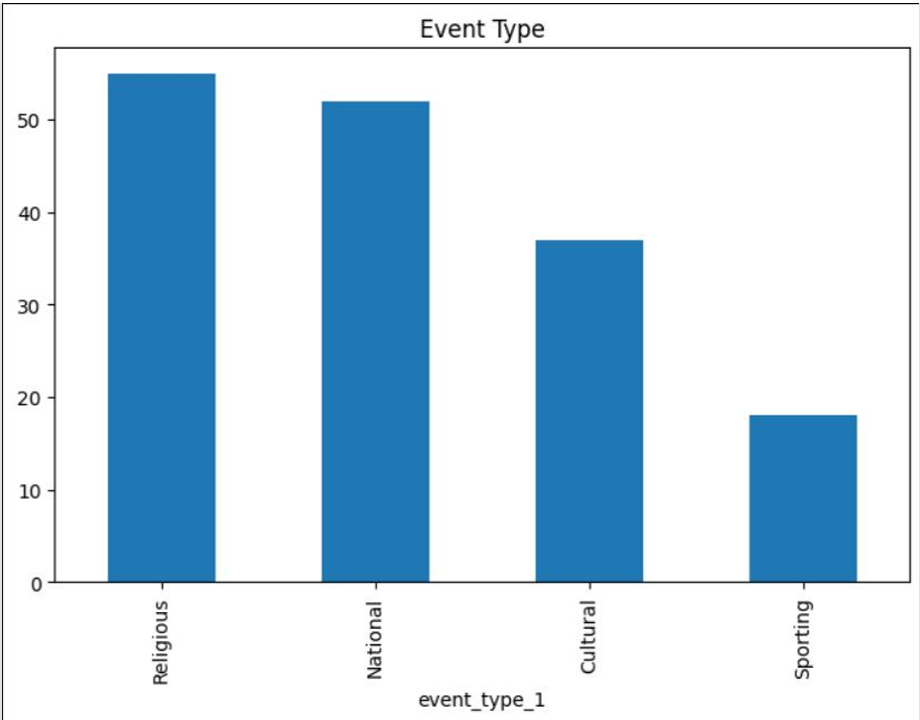
2. Weekday Distribution



Explanation:

The weekday distribution chart illustrates how sales are distributed across different days of the week. It helps identify the busiest and least active days, allowing businesses to optimize staffing, inventory, and promotional activities.

3. Event Type Distribution



Explanation:

This visualization shows the frequency of different event types recorded in the dataset. Events such as holidays and special occasions can significantly influence customer purchasing behavior. Analyzing event types helps improve forecasting accuracy by considering these seasonal effects.

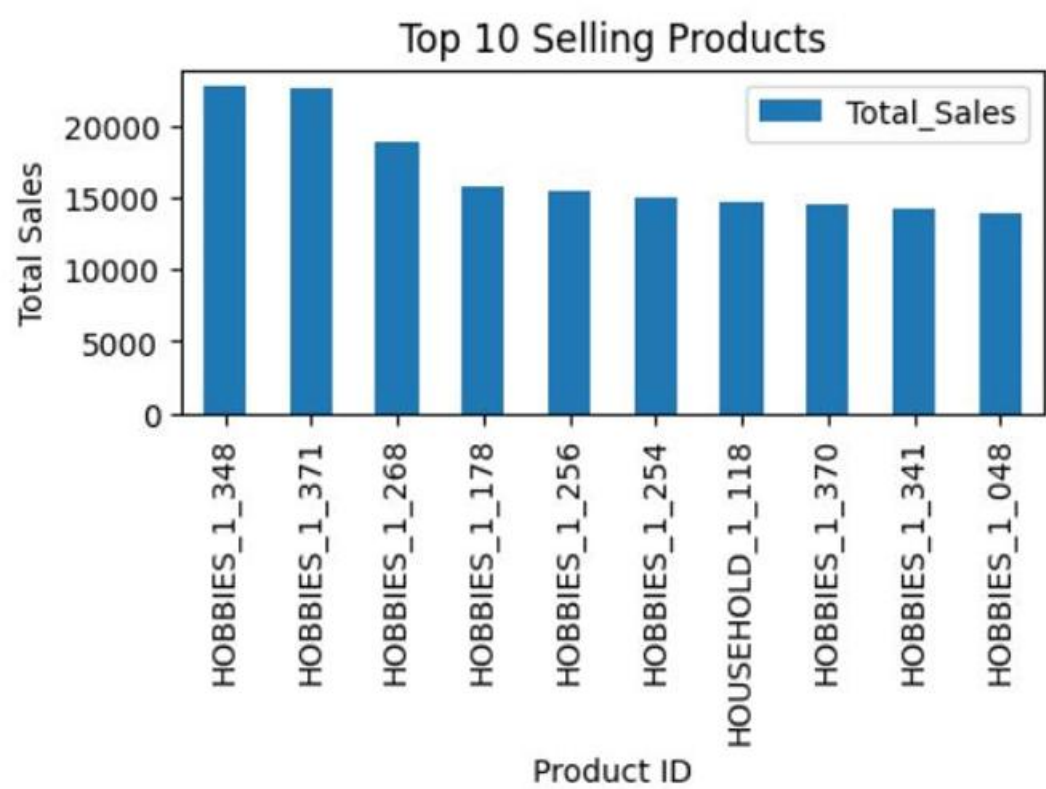
4. Sell Price Distribution



Explanation:

The sell price distribution chart represents the range and frequency of product prices. It helps identify common pricing patterns, price variability, and potential outliers, providing valuable insights into product pricing strategies.

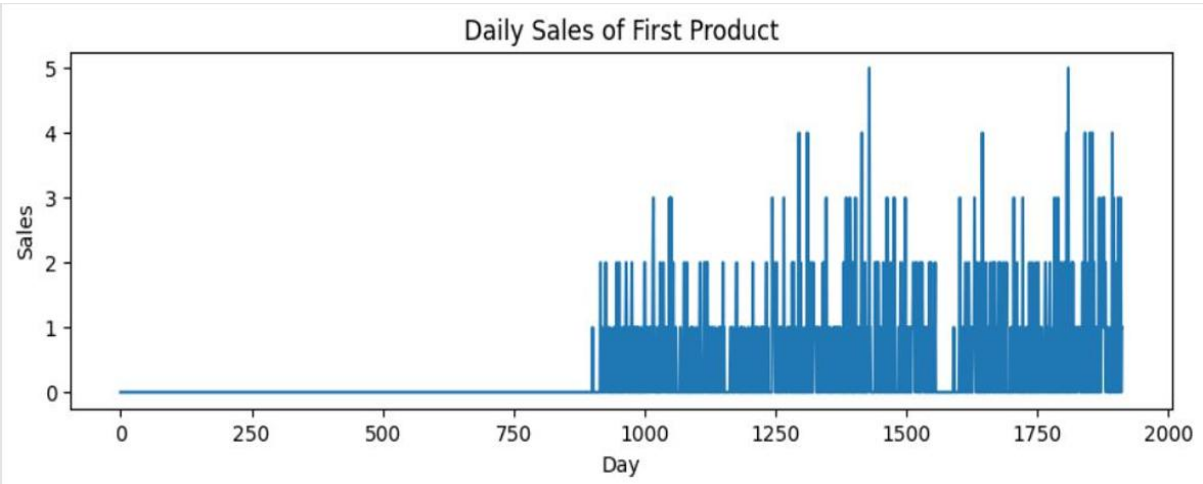
5.Top 10 Selling Products



Explanation:

This chart displays the top 10 best-selling products based on total sales. It helps identify the products with the highest customer demand. Businesses can use this information to prioritize inventory management and develop targeted marketing strategies.

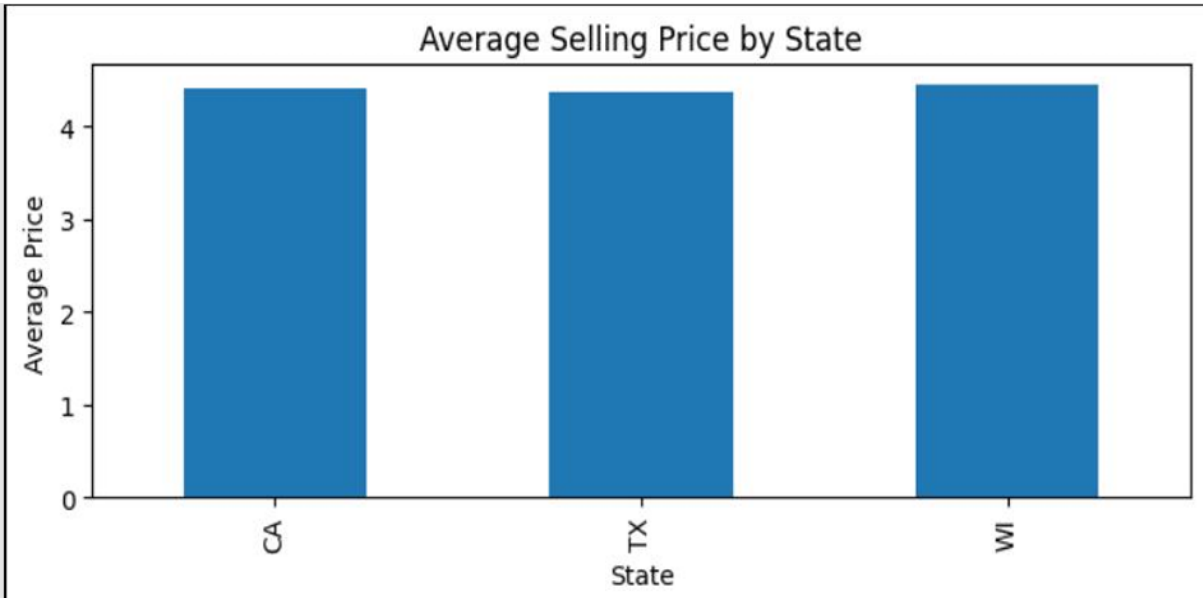
6. Daily Sales Trend



Explanation:

The daily sales trend graph illustrates how sales change over time. It helps identify periods of high and low demand, enabling businesses to understand sales patterns and improve future demand forecasting.

7. Average Selling Price by State



Explanation:

This visualization compares the average selling price of products across different states. It highlights regional pricing differences and provides insights into how product prices vary across locations.

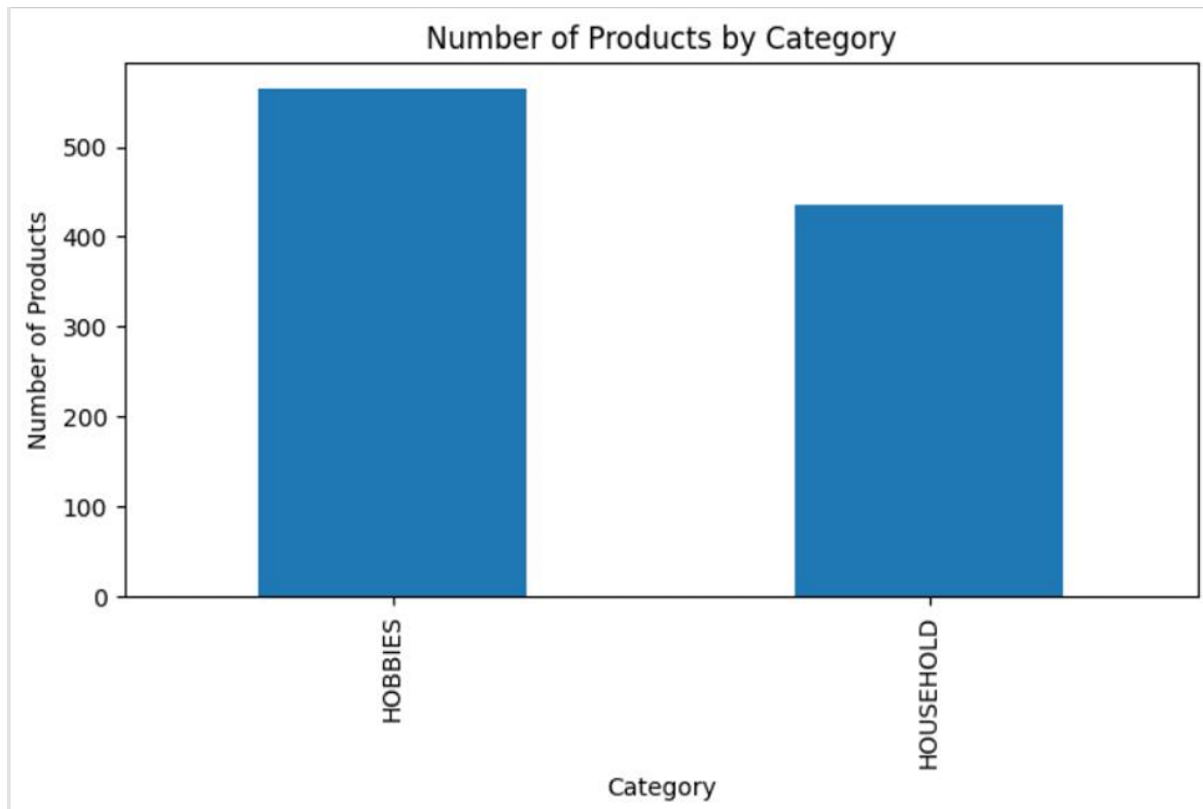
8. Average Selling Price by Store



Explanation:

The chart shows the average selling price for products in different stores. Comparing store-wise prices helps businesses evaluate pricing strategies and identify stores with higher or lower average selling prices.

9. Products by Category



Explanation:

This chart represents the distribution of products across different categories. It provides an overview of the product assortment available in the dataset and helps understand the variety of products being analyzed for demand forecasting.

Forecasting Model

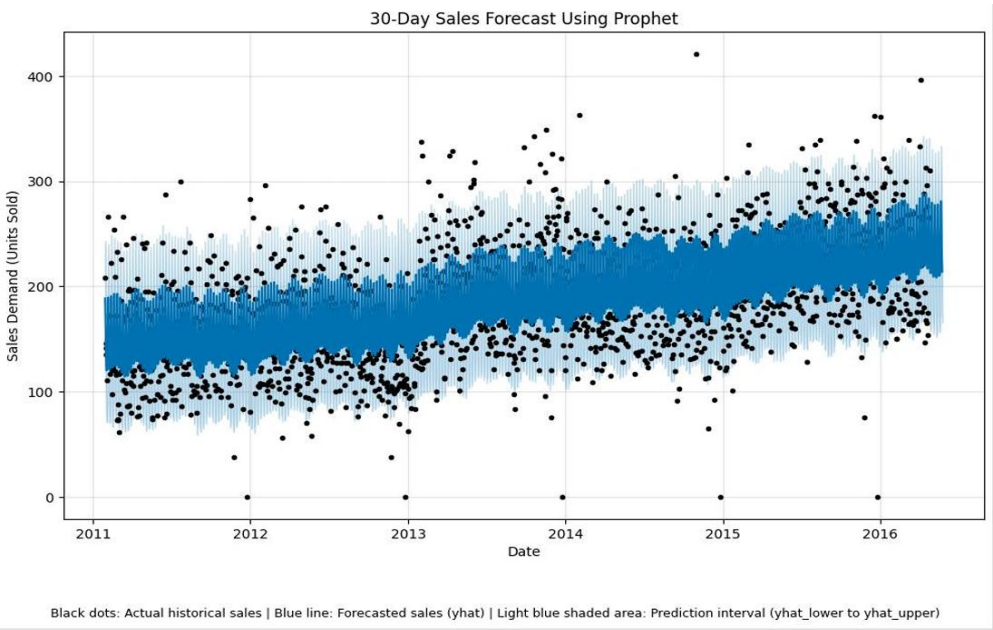
The **Facebook Prophet** model was used to forecast future retail sales. Prophet is an open-source time-series forecasting library developed by Meta (formerly Facebook). It is designed to analyze historical time-series data and generate accurate future predictions by identifying trends and seasonal patterns. Prophet is easy to implement and performs well on business forecasting problems.

The model was trained using historical retail sales data from the M5 Forecasting Dataset. After training, it generated a **30-day sales forecast**, which was used to estimate future product demand and support inventory planning.

1. Model Information

Parameter	Details
Model Name	Facebook Prophet
Algorithm Type	Time Series Forecasting
Input	Historical Retail Sales Data
Output	30-Day Sales Forecast
Purpose	Demand Forecasting for Inventory Planning

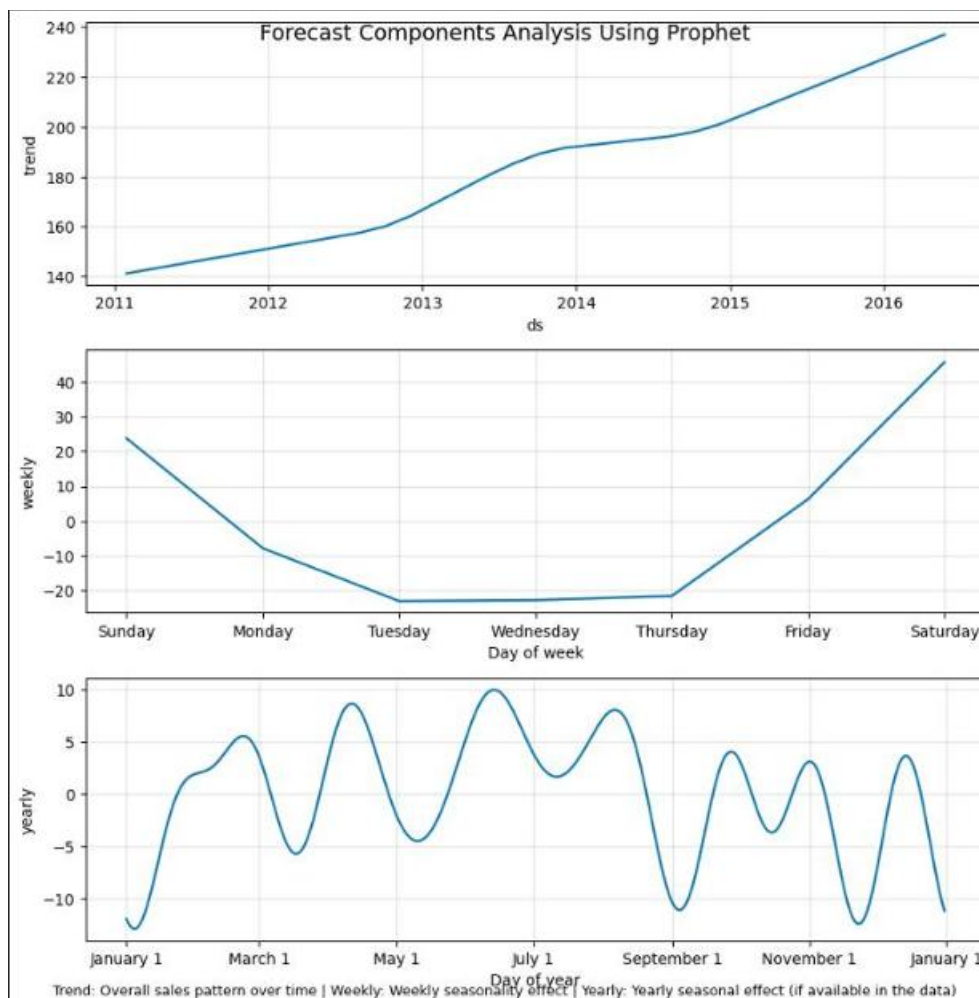
2. Sales Forecast



Explanation:

The sales forecast graph displays the predicted sales values for the next 30 days based on historical sales data. The blue line represents the forecasted sales trend, while the shaded region indicates the confidence interval of the predictions. This forecast helps businesses estimate future demand and plan inventory more effectively.

3. Forecast Components



Explanation:

The forecast components graph illustrates the individual elements that influence the predictions, such as the overall trend and seasonal patterns. Analyzing these components provides a better understanding of how the forecasting model generates future sales estimates and identifies recurring demand patterns.

Streamlit Dashboard

A user-friendly dashboard was developed using **Streamlit** to present the forecasting results in an interactive and organized manner. The dashboard allows users to view sales forecasts, analyze key metrics, and understand the forecasting model without requiring technical knowledge. It provides a simple interface for monitoring sales trends and supporting inventory planning.

1. Sales Summary

Sales Summary

Total Forecasted Sales: 356836.73

Average Predicted Sales: 183.65

Maximum Predicted Sales: 288.12

Minimum Predicted Sales: 113.53

Explanation:

The Sales Summary section displays important forecasting statistics, including total forecasted sales, average predicted sales, maximum predicted sales, and minimum predicted sales. These metrics provide a quick overview of the expected sales performance over the forecast period.

2. Dataset Information

Dataset Information

Dataset: M5 Forecasting Dataset

Forecast Records: 1943

Forecast Horizon: 30 Days

Domain: Retail Sales Forecasting

Explanation:

The Dataset Information section provides details about the dataset used in the project, including the dataset name, forecast horizon, total forecast records, and application domain. This information helps users understand the source and scope of the forecasting model.

3. Model Information

Model Information

Model: Facebook Prophet

Algorithm Type: Time Series Forecasting

Input: Historical Retail Sales Data

Output: 30-Day Sales Forecast

Purpose: Forecast future product demand to support inventory planning and business decision-making.

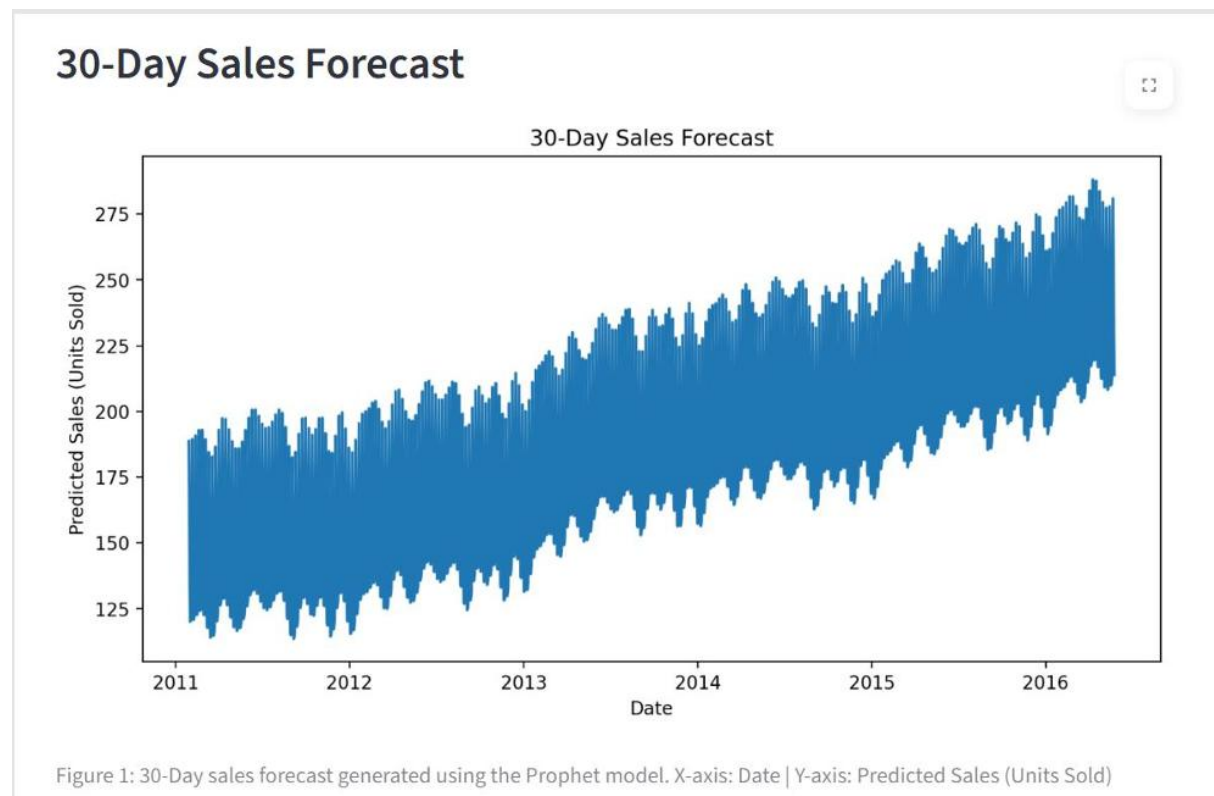
Explanation:

The Model Information section describes the forecasting model used in the project. It includes the model name (Facebook Prophet), algorithm type, input data, output, and the purpose of the model, helping users understand how the predictions are generated.

4. Inventory Recommendation**Explanation:**

This section provides recommendations based on the forecasted sales. It helps businesses estimate future inventory requirements and supports better stock management by reducing the risk of overstocking or stock shortages.

5. Forecast Results



Explanation:

The forecast results section presents the predicted sales values for the next 30 days in both graphical and tabular formats. These results enable users to analyze future demand trends and make informed business decisions.

Results and Discussion

The Retail Demand Forecasting project successfully predicted future product demand using the Facebook Prophet time-series forecasting model. After training the model with historical sales data, a **30-day sales forecast** was generated. The forecast results showed expected sales trends and seasonal variations, which can help businesses make better inventory planning decisions.

The Streamlit dashboard presented the forecast results through interactive charts and summary statistics, making it easier to understand future sales patterns. The visualizations also provided insights into product performance, pricing trends, and sales distribution across different categories and stores.

Overall, the forecasting model demonstrated its ability to estimate future demand using historical data. The project highlights how time-series forecasting can support inventory management, reduce stock shortages, and improve business planning.

1. Key Findings

- Historical sales data was successfully analyzed and prepared for forecasting.
 - The Facebook Prophet model generated a **30-day sales forecast** based on past sales trends.
 - Exploratory Data Analysis helped identify seasonal patterns, pricing trends, and product performance.
 - The Streamlit dashboard provided an interactive interface to visualize forecast results and important business metrics.
 - The forecast can assist businesses in making informed inventory and demand planning decisions.
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2. Business Impact

The forecasting system enables businesses to anticipate future demand and make proactive decisions regarding inventory management. By using forecasted sales data, organizations can reduce excess stock, minimize stock shortages, improve operational efficiency, and enhance customer satisfaction. The interactive dashboard further supports quick analysis and decision-making by presenting the forecast results in a clear and accessible format.

Advantages

The proposed retail demand forecasting system offers several benefits to businesses by improving demand prediction and inventory management. The major advantages are:

- Provides accurate demand forecasting using historical sales data.
 - Helps reduce stock shortages and excess inventory.
 - Supports better inventory planning and business decision-making.
 - Presents forecasting results through an interactive Streamlit dashboard.
 - Easy to use and understand, even for non-technical users.
 - Saves time by automating the forecasting process.
 - Assists businesses in identifying sales trends and seasonal patterns.
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Limitations

Although the project provides useful demand forecasts, it has some limitations:

- The forecasting accuracy depends on the quality and completeness of historical sales data.
 - The project uses only the Facebook Prophet model for forecasting.
 - External factors such as market trends, economic conditions, and customer preferences are not considered.
 - The application works with historical data and does not use real-time sales information.
 - Forecast results may vary if future market conditions change significantly.
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Future Scope

The project can be enhanced further by implementing the following improvements:

- Integrate real-time retail sales data for continuous forecasting.
- Compare Facebook Prophet with advanced forecasting models such as ARIMA, LSTM, or XGBoost.
- Deploy the Streamlit application on cloud platforms for online access.
- Add interactive filters and advanced visualizations to the dashboard.

- Extend forecasting to product-level and store-level demand analysis.
 - Integrate inventory optimization techniques to improve stock management.
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Conclusion

Retail demand forecasting plays a vital role in improving inventory management and supporting business decisions. In this project, the **Facebook Prophet** model was successfully used to predict retail sales for the next **30 days** using historical sales data from the **M5 Forecasting Dataset**. The developed **Streamlit dashboard** provides an interactive platform to visualize forecast results, sales summaries, and model information. Overall, the project demonstrates how time-series forecasting can help businesses anticipate future demand, optimize inventory levels, and improve operational efficiency.

References

1. M5 Forecasting Dataset (Kaggle)
2. Facebook Prophet Documentation
3. Streamlit Documentation
4. Pandas Documentation
5. NumPy Documentation
6. Matplotlib Documentation
7. Python Official Documentation